

Omar Gamal

Backend & Infrastructure Engineer | Go, Kubernetes, Distributed Systems

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github.com/Omar0Gamal | linkedin.com/in/omar-gamal-091044168 | portfolio-omar0gamals-projects.vercel.app

Remote | Fluent English | Available full-time | Can overlap with EU/US hours

Summary

Backend engineer who designed the architecture of a 14-service enterprise system (Kubernetes, Helm, Terraform) under contract and open-sourced Cartwright, a saga-based distributed checkout system, plus AxiomGraph, a C++ graph-vector database. Owns services from API design through CI/CD. Final-year CS student, building software since 2018.

Education

Capital University

B.Sc. Computer Science | GPA: 3.92 / 4.0 (Dean's List)

Expected 2027

Cairo, Egypt

Technical Skills

Languages: Go, C++, Java, Python, C#, SQL, CUDA

Backend & APIs: REST, OpenAPI, gRPC, Protocol Buffers, Gin, sqlc, ASP.NET, microservice architecture, distributed systems, role-based access control, multi-tenancy

Messaging & Data: NATS JetStream, PostgreSQL, Redis, SQLite, MongoDB, MySQL

Infrastructure: Docker, Kubernetes, Helm, Terraform, Linux, Caddy, Git, GitHub Actions, CI/CD

Observability: Prometheus, Grafana, Loki, Promtail, request tracing

Systems: Multithreading (lock striping), idempotency & retry design, CMake, vector search (HNSW)

Experience

Backend Engineer (Contract)

Sep 2025 – Nov 2025

Ayist Group

Poland (Remote)

- Designed the full architecture of a 14-service microservice system (ERP, CRM, document management), including service boundaries and inter-service communication; worked with the project manager to assign services and tasks to a team of 6 developers (Go, Python, React).
- Wrote all Protocol Buffers definitions, documented cross-service call order and failure recovery, and built the Terraform and Helm deployment for Kubernetes on Hetzner. The project was not released after the end customer withdrew.

Backend Developer & Server Administrator

Jan 2022 – Mar 2026

Attila Gaming

United States (Remote)

- Built matchmaking, game-instance spawning/lifecycle management, and anti-cheat features across 15+ Java/Spigot plugins for a Minecraft server with 500+ concurrent players.
- Administer the Linux servers (CI/CD, containerized services, automated backups); reduced crashes by 60% and improved response times by 40%.

Projects

Cartwright | Go, C#/ASP.NET, Python, gRPC, NATS, K8s

github.com/Omar0Gamal/cartwright-saga

- Saga-based distributed order-checkout system: Go orchestrator, C#/ASP.NET billing service, Python notifier over gRPC/Protocol Buffers; compensating transactions and a transactional outbox to NATS JetStream.
- Idempotency keys, backoff retries, and lease-based recovery; a scripted chaos test kills the billing service mid-request and confirms no double charge on recovery. Deployed via Helm to k3s provisioned with Terraform; CI across all three languages.

AxiomGraph | C++17, CUDA, Python, SQLite, CMake

github.com/Omar0Gamal/AxiomGraph

- Hybrid graph-vector database in C++17 for GraphRAG: lock-striped graph consolidated to CSR, HNSW (CPU) and NVIDIA cuVS (GPU) indices, SQLite metadata store; nanobind Python SDK, HTTP server, CI-built wheels.
- 68.4% All-Gold-Found@10 on HotpotQA bridge questions (vs. 45.8% dense + rerank); ~8,600 QPS (CPU) / 33,000+ QPS (GPU); 1M nodes in ~2 GB RAM.

Open Source

llama.cpp – RPC server bounds-check fix | C++

github.com/ggml-org/llama.cpp/pull/26933 (open)

- Fixed two reported out-of-bounds memory bugs (GET_ROWS read, SET_ROWS write) in llama.cpp's RPC server by validating tensor shapes before graph execution. Pull request open, awaiting review.